nature portfolio

Peer Review File

Efficient Pt1Ni single-atom alloy catalyst for hydrogen-free catalytic fractionation of lignocellulose

Corresponding Author: Professor Yanqin Wang

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
This work reports that a single-atom alloy Pt1Ni catalyst can directly convert birch sawdust into propenyl side-chained phenols with a selectivity of 50%. The study is significant and has substantial potential impact; however, several aspects require revision prior to publication, particularly the mechanistic investigation.
Major comments that need to be addressed include:
1. From the activity data in Table 1, comparison of entries 3 and 4 suggests that the presence of Ni species has only a minor effect on overall reaction yield. However, when comparing entries 1 and 7, introduction of Ni increases the yield from 0 to 38.6%. These observations indicate that Ni primarily enhances catalytic activity rather than selectively steering formation of propenyl side-chained phenols. The manuscript should present a clear, quantitative statement of this conclusion, and

relationship.
4. The second paragraph of the introduction should be more concise and precise. If the key advancement of this work is the high selectivity toward propenyl side-chained phenols, the discussion should focus primarily on this point and avoid unrelated or secondary descriptions.
5. In the first paragraph of the Results and Discussion section, a more detailed introduction of the catalyst systems is needed. Rather than only listing labels such as “2Pt/NiAl2O4” or “5Ni/NiAl2O4,” please briefly describe their compositions, structural features, and why they were selected for comparison.
6. The current explanation linking PtOx reduction peaks in the H2-TPR profile to Pt dispersion is not scientifically accurate. The reduction behavior of PtOx does not directly reflect Pt dispersion. Additional clarification and a more appropriate interpretation of the TPR results are required.
7. The manuscript should explicitly state what conclusions can be drawn from the H2-TPR experiments.

Reviewer #2

(Remarks to the Author)
The submitted manuscript reports a single-atom alloy Pt_1Ni catalyst that enables reductive catalytic fractionation (RCF) of birch sawdust, achieving a phenolic monomer yield of 50.9 wt.% with ~50% selectivity toward valuable propenyl-substituted products under relatively mild conditions (140 °C, 1 atm N_2), while largely preserving cellulose. Although the reported monomer yield is somewhat higher than that in several previous RCF studies, the catalyst design and mechanistic insights do not appear to offer sufficient novelty or depth relative to the existing literature on single-atom and bimetallic RCF catalysts. The advances claimed are therefore incremental rather than transformative. Consequently, I recommend rejection of the manuscript in its present form. The following comments are provided to help improve the work for future submission.

1. Both $0.2Pt5Ni/NiAl_2O_4$ and $2Pt5Ni/NiAl_2O_4$ exhibit hydrogen-free lignin fractionation at 140 °C, which is significantly lower than the temperatures required for the catalysts shown in Fig. S6, whereas bare $NiAl_2O_4$ is inactive under identical conditions. This raises an important mechanistic question regarding the specific role of the $NiAl_2O_4$ support in enabling this unusually low-temperature activity. The authors should more clearly elucidate whether the support contributes through metal–support interactions, hydrogen spillover, oxygen vacancies, or promotion of lignin depolymerization pathways.
2. The manuscript argues that Ni is not directly the active site, yet its oxophilicity is essential for promoting Cα–OH hydrogenolysis and the formation of propenyl side chains. If oxophilicity is indeed the dominant factor, the authors should rationalize why other more strongly oxophilic transition metals (e.g., V, Nb, Mo, W) were not considered or screened. A comparative discussion or preliminary screening would significantly strengthen the mechanistic argument.
3. The reported turnover number (TON) of 270.5 for $0.2Pt5Ni/NiAl_2O_4$ is only moderate compared to recently reported Ru-based single-atom catalysts for lignin RCF. Moreover, the long reaction time (12 h) required to achieve the high monomer yield suggests a relatively low intrinsic reaction rate. The authors should demonstrate that the improved monomer yield originates from genuinely enhanced site activity rather than simply from extended reaction time. Time-resolved kinetics, initial rate comparisons, or space–time yield (STY) analysis would be necessary to support this claim.
4. Additional control experiments are required to verify the role of xylan as the hydrogen source. For example, reactions conducted under H_2 or using external hydrogen-donating solvents should be included to determine whether the formation of propenyl-substituted phenolic products is dictated primarily by the hydrogen source rather than by the intrinsic catalytic properties. Furthermore, the selectivity toward propenyl side-chained products should be quantitatively compared with literature values to convincingly demonstrate any advantage of single-atom alloy catalysts over conventional single-atom or nanoparticle catalysts for preserving unsaturated side chains.
5. To enhance the reliability of the results, more experimental details are needed. For example, monomer yield quantitative method, GC column temperature program, representative GC chromatograms with peak assignments，GPC results for the lignin-oil.

Reviewer #3

(Remarks to the Author)
Zhou et al. report the development of a single-atom alloyed Pt_1Ni catalyst for lignocellulose fractionation that enables efficient production of phenolic monomers under hydrogen-free conditions while preserving cellulose integrity. This work addresses a critical challenge in biomass valorization by achieving high monomer yields and enhanced selectivity toward unsaturated side-chain products without the need for external hydrogen or excessive metal loading, and it proposes a mechanistically well-supported catalytic strategy. Overall, the manuscript is strong and appears suitable for publication. However, several concerns should be addressed, as outlined below. Addressing these points will further strengthen the manuscript and ensure that it represents a meaningful contribution to sustainable biomass conversion and catalytic lignin valorization.

5. The manuscript contains some typos. For example, “Tranditionally” should be corrected to “Traditionally” in page3, and “efficientcy” should be corrected to “efficiency” in page3 and page12. The authors are encouraged to carefully proofread the text and revise these issues for clarity and consistency.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Many thanks to the authors. Now the current version is more appropriate for publication.

Reviewer #2

(Remarks to the Author)
The manuscript provides detailed explanations for the proposed issues and supplements relevant experiments. There are still some questions that require to provide further clarification before publication.

1) The manuscript suggests that Al_2O_3 and $NiAl_2O_4$ have different reaction pathways, which is driven by abundant oxygen vacancies, promoted hydrogen spillover, and robust metal-support interactions. The authors should further elaborate on the fundamental reasons underlying these observed differences.
2) Aerobic metals Mo and W cannot form alloy structures with Pt, but still exhibit SCF activity and a certain selectivity for propenyl products compared to 0.2% Pt. Please provide a more detailed explanation to clarify this.

Reviewer #3

(Remarks to the Author)
My questions have been well addressed. I recommend the acceptance of the revised manuscrupt as is.

Response to referees' comments and questions

Reviewer #1 (Remarks to the Author):

This work reports that a single-atom alloy Pt1Ni catalyst can directly convert birch sawdust into propenyl side-chained phenols with a selectivity of 50%. The study is significant and has substantial potential impact; however, several aspects require revision prior to publication, particularly the mechanistic investigation.

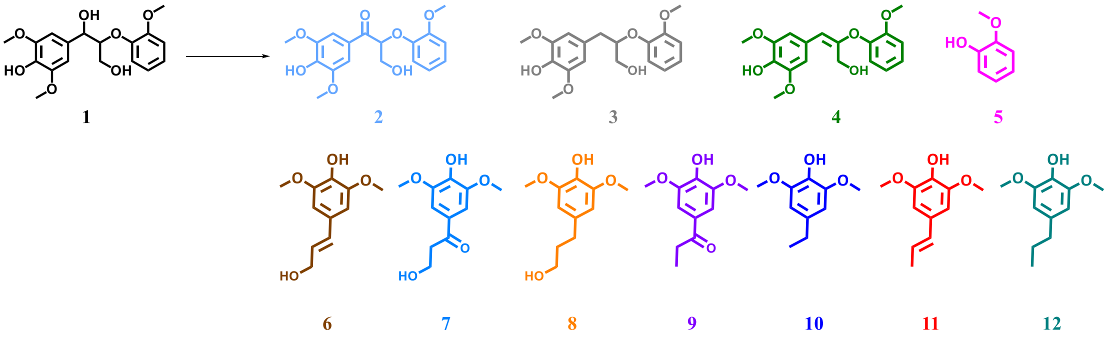
Major comments that need to be addressed include:

1. From the activity data in Table 1, comparison of entries 3 and 4 suggests that the presence of Ni species has only a minor effect on overall reaction yield. However, when comparing entries 1 and 7, introduction of Ni increases the yield from 0 to 38.6%. These observations indicate that Ni primarily enhances catalytic activity rather than selectively steering formation of propenyl side-chained phenols. The manuscript should present a clear, quantitative statement of this conclusion, and support it with additional experimental evidence.

Reply: We thank the reviewer for raising this comment. Metal Ni plays a unique role in the single-atom alloy Pt_1Ni catalyst. The addition of Ni forms a highly active Pt_1-Ni single-atom alloy structure, which successfully depolymerizes lignin to yield a 50.9 wt% yield of phenolic monomers with about 50% selectivity to propenyl side-chained products. Furthermore, compared to 0.2Pt catalyst, the oxygen affinity of Ni sites facilitates the formation of propenyl side-chained phenols on Pt_1-Ni catalyst. To visualize this effect, we compared the depolymerization activity of 0.2Pt5Ni and the 0.2Pt catalysts toward lignin model compound, syringyl-glycerol-β-guaiacyl ether (**SG**, as shown in Table R1) by using xylan as the hydrogen source. The results indicate that intermediate 4 is the primary product on the 0.2Pt5Ni catalyst, yielding 28.5% of the olefinic phenolic monomer (6+11). In contrast, on the 0.2Pt catalyst, intermediates 2 and 4 predominated, with a 13.7% yield of olefinic phenolic monomers originating from 6. This indicates that 0.2Pt5Ni and 0.2Pt catalysts exhibit distinct reaction pathways, with the former producing more propenyl side-chained phenols via intermediate 4. In addition, the 0.2Pt5Ni catalyst demonstrates superior hydrogen production capacity compared to the 0.2Pt catalyst (Table R2), which could accelerate lignin depolymerization.

These results further demonstrate the crucial role of Ni in enhancing catalytic activity and formation of propenyl side-chained phenols. The relevant description has been incorporated into the revised manuscript and supporting information (Tables S3 and S4).

Table R1. Reaction pathway of lignin model compound **SG** over two different catalysts.



Entry	Catalysts	Time (h)	Conv. (%)	Yield (wt%)										
				2	3	4	5	6	7	8	9	10	11	12
1	0.2Pt5Ni	2	100	0.4	0.1	7.6	37.1	12.0	4.5	0.1	0.7	1.6	16.5	2.7
2	0.2Pt	2	98.2	21.3	-	14.8	26.0	13.7	5.6	0.2	0.4	-	0.3	0.9

Reaction Conditions: Xylan (0.05 g), Substrate (0.1 g), catalyst (0.05 g), H_2O (10 mL), N_2 1 atm, 140 °C.

Table R2. Hydrogen production of xylan over different catalysts.

Catalysts	H_2 Content/μmol·g^{-1}
0.2Pt5Ni	502
0.2Pt	261
5Ni	103
2Pt	1113
2Pt5Ni	1232

Reaction Conditions: Xylan (0.2 g), Catalyst (0.1 g), H_2O (10 mL), N_2 1 atm, 140 °C, 2 h.

Other comments:

3. Reductive catalytic fractionation (RCF) is not equivalent to the lignin-first biorefining strategy. Please

revise the introduction to clearly distinguish these two concepts and provide a more accurate and coherent summary of their relationship.

Reply: We thank the reviewer for this suggestion. We agree that treating RCF as synonymous with the broader lignin-first strategy is inaccurate. We recognize that lignin-first is an overarching strategy defined by the active stabilization of lignin during fractionation (including RCF, oxidative catalytic fractionation (OCF), and protection-group approaches), whereas RCF is a specific methodology relying on reductive catalysis.

In the revised manuscript, we have rewritten the relevant paragraph in the introduction to explicitly clarify this hierarchical relationship. The rewritten content is as follows:

"The lignin-first biorefining has emerged as a pivotal strategy. This overarching strategy facilitates the active stabilization of reactive lignin intermediates during biomass fractionation to prevent condensation reactions. Broadly, the lignin-first encompasses various stabilization methods including reductive catalytic fractionation (RCF),[1,2] oxidative catalytic fractionation (OCF),[3,4] protection-group approaches.[5] Among these methods, RCF is widely used in biomass fractionation due to the high lignin phenolic monomers yield and good (hemi)cellulose retention.[6,7]"

1. Van den Bosch, S., *et al.* Reductive lignocellulose fractionation into soluble lignin-derived phenolic monomers and dimers and processable carbohydrate pulps. *Energy Environ. Sci.* **8**, 1748-1763 (2015).
2. Li, N., *et al.* Selective lignin arylation for biomass fractionation and benign bisphenols. *Nature* **630**, 381-386 (2024).
3. Luo, H., *et al.* Oxidative Catalytic Fractionation of Lignocellulosic Biomass under Non-alkaline Conditions. *J. Am. Chem. Soc.* **143**, 15462-15470 (2021).
4. Zhu, Y., *et al.* Oxidative Catalytic Fractionation of Lignocellulose to High-Yield Aromatic Aldehyde Monomers and Pure Cellulose. *ACS Catal.* **13**, 7929-7941 (2023).
5. Shuai, L., *et al.* Formaldehyde stabilization facilitates lignin monomer production during biomass depolymerization. *Science* **354**, 329-333 (2016).
6. Abu-Omar, M.M., *et al.* Guidelines for performing lignin-first biorefining. *Energy Environ. Sci.* **14**, 262-292 (2021).
7. Renders, T., Van den Bosch, S., Koelewijn, S.-F., Schutyser, W. & Sels, B. Lignin-first biomass fractionation: the advent of active stabilisation strategies. *Energy Environ. Sci.* **10**, 1551-1557 (2017).

4. The second paragraph of the introduction should be more concise and precise. If the key advancement of this work is the high selectivity toward propenyl side-chained phenols, the discussion should focus primarily

on this point and avoid unrelated or secondary descriptions.

Reply: We thank the reviewer for this advice. We have restructured the language in the second paragraph to focus on the discussion of propenyl side-chained phenolic monomers, streamlining secondary points. The revised paragraph is as follows:

"The product selectivity of RCF is highly tunable based on the catalytic system,[8-14] typically favoring propyl-, propanol-, or propenyl-substituted phenols. Of these, the propenyl variants have garnered significant attention due to their unique double bonds, which enable functionalization into various high-value chemicals.[15] Samec et al. achieved 49 wt% yield of 2,6-dimethoxy-4-(prop-1-enyl)-phenol from birch wood using ethanol/water as solvent over 5 wt%Pd/C catalyst.[16] In addition, a choline chloride/ethylene glycol-based system was developed for hydrogen transfer reduction catalytic fractionation, yielding 36.2% propenyl phenols with 87.6% selectivity, but resulting in partial degradation of carbohydrates.[17] Besides, catalysts with low loading and high dispersion also demonstrate outstanding performance in RCF. Song and coworkers reported the atomic dispersed ruthenium catalyst (Ru/ZnO/C) and palladium catalyst (Pd/ZnO/C) for acquiring propenyl-catechol (selectivity of 77%) and propenyl-phenols (selectivity of 62%), respectively.[18,19]"

8. Zhai, Y., *et al.* Depolymerization of lignin via a non-precious Ni–Fe alloy catalyst supported on activated carbon. *Green Chem.* **19**, 1895-1903 (2017).
9. Park, J., *et al.* Highly efficient reductive catalytic fractionation of lignocellulosic biomass over extremely low-loaded Pd catalysts. *ACS Catal.* **10**, 12487-12506 (2020).
10. Liu, Z., *et al.* Rational highly dispersed ruthenium for reductive catalytic fractionation of lignocellulose. *Nat. Commun.* **13**, 4716 (2022).
11. Ge, J., *et al.* Highly efficient metal-acid synergetic catalytic fractionation of lignocellulose under mild conditions over lignin-coordinated N-anchoring Co single-atom catalyst. *Chem. Eng. J.* **462**, 142109 (2023).
12. Brienza, F., *et al.* Toward a Hydrogen-Free Reductive Catalytic Fractionation of Wheat Straw Biomass. *ChemSusChem* **16**, e202300103 (2023).
13. Chen, J., *et al.* Efficient fractionation and catalytic valorization of raw biomass in ϵ-caprolactone and water. *ChemSusChem* **16**, e202202162 (2023).
14. Zhang, K., *et al.* Catalytic Hydrogenolysis of Lignin into Propenyl-monophenol over Ru Single Atoms Supported on CeO2 with Rich Oxygen Vacancies. *ACS Catal.* **14**, 16115-16126 (2024).
15. Trullemans, L., *et al.* Renewable and safer bisphenol A substitutes enabled by selective zeolite alkylation. *Nat. Sustain.* **6**, 1693-1704 (2023).
16. Galkin, M.V. & Samec, J.S. Selective route to 2-propenyl aryls directly from wood by a tandem organosolv and palladium-catalysed transfer hydrogenolysis. *ChemSusChem* **7**, 2154-2158 (2014).
17. Li, Y., *et al.* Hydrogen-Transfer Reductive Catalytic Fractionation of Lignocellulose: High Monomeric Yield with Switchable Selectivity. *Angew. Chem. Int. Ed.* **62**, e202307116 (2023).
18. Wang, S., Zhang, K., Li, H., Xiao, L.P. & Song, G. Selective hydrogenolysis of catechyl lignin into

propenylcatechol over an atomically dispersed ruthenium catalyst. *Nat. Commun.* **12**, 416 (2021).
19. Wang, S., Li, X., Fu, C., Li, H. & Song, G. Atomically Dispersed Palladium Driving Reductive Catalytic Fractionation of Lignocellulose into Alkene-Functionalized Phenols. *ACS Catal.* **14**, 3565-3574 (2024).

5. In the first paragraph of the Results and Discussion section, a more detailed introduction of the catalyst systems is needed. Rather than only listing labels such as "2Pt/NiAl2O4" or "5Ni/NiAl2O4," please briefly describe their compositions, structural features, and why they were selected for comparison.

Reply: We thank the reviewer for raising this comment. As comparison, $2Pt/NiAl_2O_4$ and $5Ni/NiAl_2O_4$ catalysts were also prepared with the same method. The former (monometallic Pt loaded on spinel) was selected as a high-performance benchmark based on our previous work,[20] and the latter (monometallic Ni loaded on spinel) was chosen to determine the intrinsic baseline activity of pure Ni catalyst.

The above content has been incorporated into the revised manuscript.

20. Zhou, H., Liu, X., Guo, Y. & Wang, Y. Self-hydrogen supplied catalytic fractionation of raw biomass into lignin-derived phenolic monomers and cellulose-rich pulps. *JACS Au* **3**, 1911-1917 (2023).

6. The current explanation linking PtOx reduction peaks in the H2-TPR profile to Pt dispersion is not scientifically accurate. The reduction behavior of PtOx does not directly reflect Pt dispersion. Additional clarification and a more appropriate interpretation of the TPR results are required.

Reply: We thank the reviewer for raising this comment. We have removed the correlation between PtO_x reduction behavior and Pt dispersion. The reduction peaks of PtO_x were not detected for the 0.2Pt5Ni catalyst. This is likely due to the low loading of Pt (0.2 wt%), which falls below the detection limit of the TCD.

The above content has been added into the revised manuscript.

7. The manuscript should explicitly state what conclusions can be drawn from the H2-TPR experiments.

Reply: We thank the reviewer for raising this comment. We have incorporated the H_2-TPR conclusions into the revised manuscript as follows:

"Specifically, the H_2-TPR profiles reveal that Pt facilitates NiO reduction via hydrogen spillover, where dissociated hydrogen on Pt reduces adjacent NiO species. This confirms the structural intimacy of Pt and Ni, supporting potential alloy formation and enhancing overall catalyst reducibility."

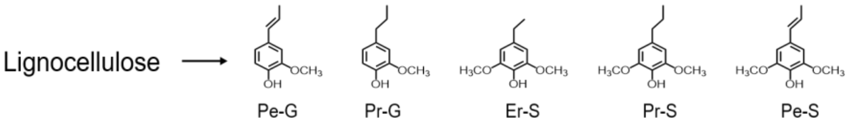
Reviewer #2 (Remarks to the Author):

The submitted manuscript reports a single-atom alloy Pt_1Ni catalyst that enables reductive catalytic fractionation (RCF) of birch sawdust, achieving a phenolic monomer yield of 50.9 wt.% with ~50% selectivity toward valuable propenyl-substituted products under relatively mild conditions (140 °C, 1 atm N_2), while largely preserving cellulose. Although the reported monomer yield is somewhat higher than that in several previous RCF studies, the catalyst design and mechanistic insights do not appear to offer sufficient novelty or depth relative to the existing literature on single-atom and bimetallic RCF catalysts. The advances claimed are therefore incremental rather than transformative. Consequently, I recommend rejection of the manuscript in its present form. The following comments are provided to help improve the work for future submission.

1. Both $0.2Pt5Ni/NiAl_2O_4$ and $2Pt5Ni/NiAl_2O_4$ exhibit hydrogen-free lignin fractionation at 140 °C, which is significantly lower than the temperatures required for the catalysts shown in Fig. S6, whereas bare $NiAl_2O_4$ is inactive under identical conditions. This raises an important mechanistic question regarding the specific role of the $NiAl_2O_4$ support in enabling this unusually low-temperature activity. The authors should more clearly elucidate whether the support contributes through metal–support interactions, hydrogen spillover, oxygen vacancies, or promotion of lignin depolymerization pathways.

Reply: We thank the reviewer for raising this comment. To better elucidate the unique role of the $NiAl_2O_4$ support in the reaction, we synthesized a $0.2Pt5Ni/Al_2O_3$ catalyst using Al_2O_3 as the support for comparison. In the self-hydrogen supplied fractionation of birch wood (Table R3), only 7.3 % yield of phenolic monomers was achieved over $0.2Pt5Ni/Al_2O_3$ catalyst, significantly lower than 38.6 % obtained over $0.2Pt5Ni/NiAl_2O_4$ catalyst. In the reaction of model compound **SG** (Table R4), the intermediate 4 is the primary product on the $0.2Pt5Ni/NiAl_2O_4$ catalyst, yielding 28.5% of the olefinic phenolic monomer (6+11). In contrast, on the $0.2Pt5Ni/Al_2O_3$ catalyst, intermediates 2 and 4 predominated, with a 18% yield of ketone phenolic monomers (7+9). This indicates that different reaction pathways are exhibited on $NiAl_2O_4$ and Al_2O_3 supports.

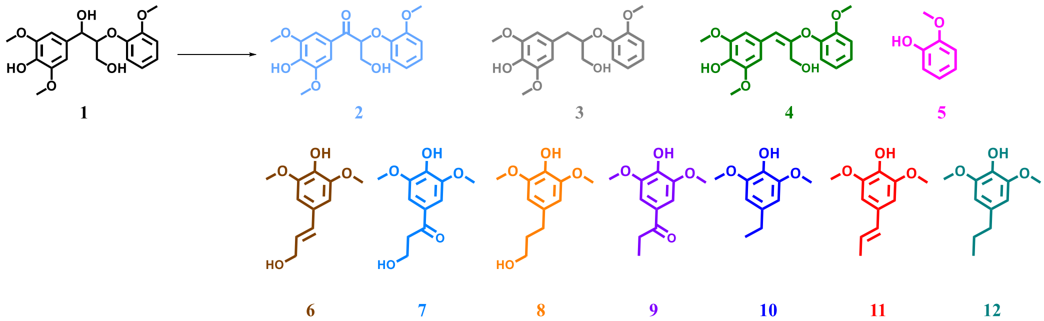
Table R3. SCF of birch wood over different catalysts.



Catalysts	Phenolic monomers yield (wt%)					Sel. to Pe-G/S (%)	Total (wt%)
	Pe-G	Pr-G	Er-S	Pr-S	Pe-S		
0.2Pt5Ni/NiAl$_2$O$_4$	0.5	1.2	2.4	15.0	19.5	51.8	38.6
0.2Pt5Ni/Al$_2$O$_3$	-	-	-	3.4	3.9	53.4	7.3

Reaction conditions: birch wood (0.5 g), catalyst (0.2 g), H_2O (10 mL), N_2 1 atm, 140 °C, 12 h.

Table R4. Reaction pathway of lignin model compound **SG** over two different catalysts.



Catalysts	Conv. (%)	Yield (wt%)										
		2	3	4	5	6	7	8	9	10	11	12
0.2Pt5Ni/NiAl$_2$O$_4$	100	0.4	0.1	7.6	37.1	12.0	4.5	0.1	0.7	1.6	16.5	2.7
0.2Pt5Ni/Al$_2$O$_3$	100	9.2	-	13.1	32.5	5.7	15.2	0.4	2.8	0.3	5.2	3.7

Reaction Conditions: Xylan (0.05 g), Substrate (0.1 g), catalyst (0.05 g), H_2O (10 mL), N_2 1 atm, 140 °C, 2 h.

Furthermore, we conducted H_2-TPR, electron paramagnetic resonance (EPR), and hydrogen spillover experiments over 0.2Pt5Ni/NiAl$_2$O$_4$ and 0.2Pt5Ni/Al$_2$O$_3$ (Figures R1-R3) to investigate the influence of the support on the properties of the catalysts. The 0.2Pt5Ni/NiAl$_2$O$_4$ catalyst exhibits a higher number of oxygen vacancies (Figure R1) and stronger hydrogen spillover capacity compared to the 0.2Pt5Ni/Al$_2$O$_3$ catalyst (Figure R2). In addition, in the H_2-TPR profiles (Figure R3), the co-reduction peak of PtO_x and NiO_x species on 0.2Pt5Ni/NiAl$_2$O$_4$ catalyst appears at a higher temperature (371° C), indicating stronger metal-support interactions on 0.2Pt5Ni/NiAl$_2$O$_4$ catalyst than that of 0.2Pt5Ni/Al$_2$O$_3$ catalyst (360° C).

In summary, the 0.2Pt5Ni/$NiAl_2O_4$ catalyst exhibits superior performance in the selective depolymerization of lignin to propenyl side-chained phenolic monomers at 140° C. This is driven by abundant oxygen vacancies, promoted hydrogen spillover, and robust metal-support interactions. Notably, the superior hydrothermal stability of the $NiAl_2O_4$ support relative to Al_2O_3 support plays an indispensable role in this process.

The relevant description has been incorporated into the revised manuscript and supporting information (Tables S7 and S8, Figures S11-S13).

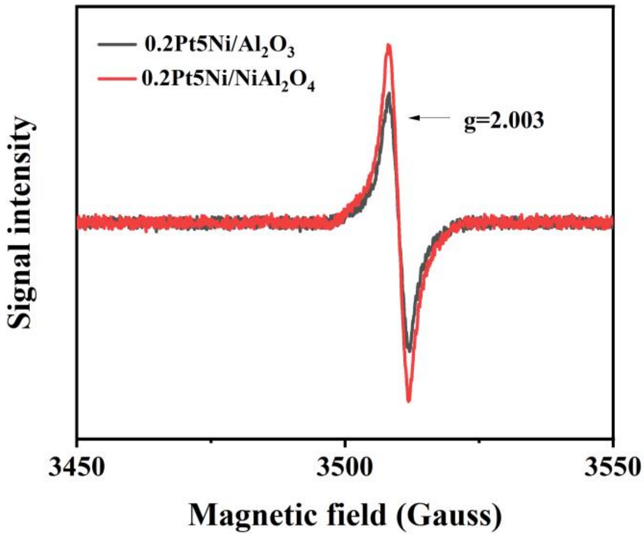


Figure R1. EPR profiles of 0.2Pt5Ni/$NiAl_2O_4$ and 0.2Pt5Ni/Al_2O_3 catalysts.

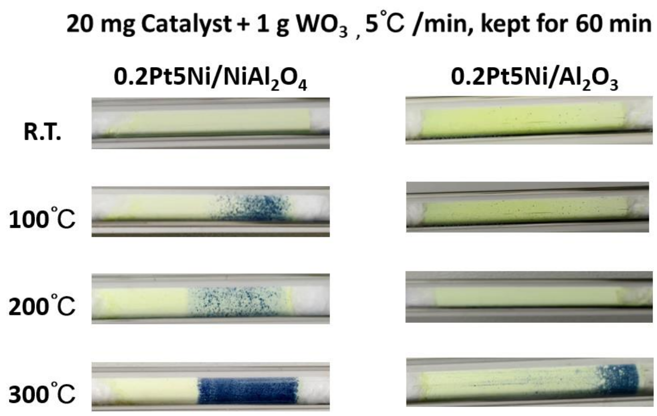


Figure R2. Hydrogen spillover experiments over 0.2Pt5Ni/$NiAl_2O_4$ and 0.2Pt5Ni/Al_2O_3 catalysts.

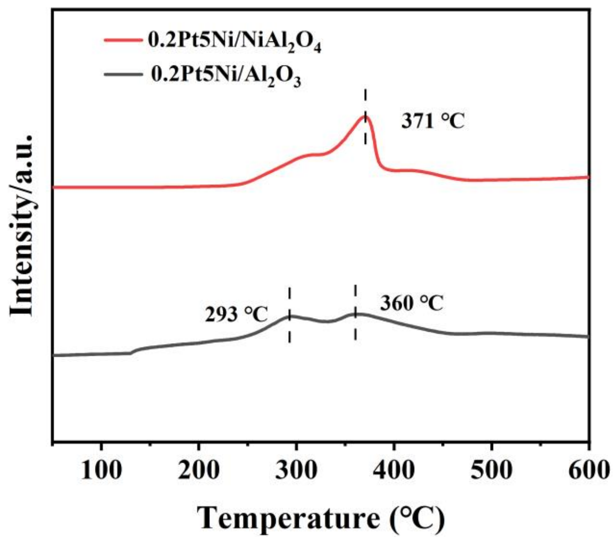


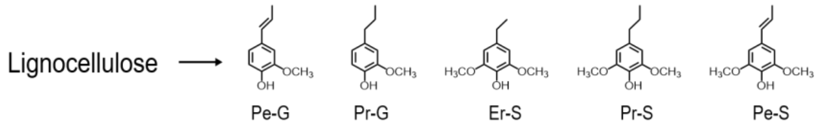
Figure R3. H_2-TPR profiles of 0.2Pt5Ni/$NiAl_2O_4$ and 0.2Pt5Ni/Al_2O_3 catalysts.

2. The manuscript argues that Ni is not directly the active site, yet its oxophilicity is essential for promoting Cα–OH hydrogenolysis and the formation of propenyl side chains. If oxophilicity is indeed the dominant factor, the authors should rationalize why other more strongly oxophilic transition metals (e.g., V, Nb, Mo, W) were not considered or screened. A comparative discussion or preliminary screening would significantly strengthen the mechanistic argument.

Reply: We thank the reviewer for this advice. We supplemented the activity of various oxygen affinity transition metals (e.g., Nb, Mo, W) in self-hydrogen supplied fractionation of birch wood as shown in Table R5. Results indicate that the 0.2Pt5Ni catalyst exhibits the highest activity with a 38.6% phenolic monomers yield, whereas the 0.2Pt5W and 0.2Pt5Mo catalysts yield only 14.7% and 8.9% phenolic monomers, respectively. Notably, no phenolic monomers were detected on 0.2Pt5Nb catalyst. Instead, significant furfural products were observed in the post-reaction solution, indicating that the strong acidity from Nb converted hemicellulose into furfural in the lignin depolymerization process. The reason for the above results may be that Nb, Mo, and W are difficult to be reduced and cannot form alloy structures with metal Pt, further demonstrating the crucial role of Ni in self-hydrogen supplied fractionation.

The relevant description has been incorporated into the revised manuscript and supporting information (Table S9).

Table R5. SCF of birch wood over four different catalysts.



Entry	Sample	Phenolic monomers yield (wt%)					Sel. to Pe-G/S (%)	Total (wt%)
		Pe-G	Pr-G	Er-S	Pr-S	Pe-S		
1	0.2Pt5Ni	0.5	1.2	2.4	15.0	19.5	51.8	38.6
2	0.2Pt5Nb	-	-	-	-	-	-	-
3	0.2Pt5W	-	-	0.2	5.6	8.9	60.5	14.7
4	0.2Pt5Mo	-	-	0.4	4.1	4.4	49.4	8.9

Reaction conditions: substrate (0.5 g), catalyst (0.2 g), H_2O (10 mL), N_2 1 atm, 140 °C, 12 h.

3. The reported turnover number (TON) of 270.5 for 0.2Pt5Ni/$NiAl_2O_4$ is only moderate compared to recently reported Ru-based single-atom catalysts for lignin RCF. Moreover, the long reaction time (12 h) required to achieve the high monomer yield suggests a relatively low intrinsic reaction rate. The authors should demonstrate that the improved monomer yield originates from genuinely enhanced site activity rather than simply from extended reaction time. Time-resolved kinetics, initial rate comparisons, or space–time yield (STY) analysis would be necessary to support this claim.

Reply: We thank the reviewer for raising this comment. To address the concern regarding "reaction time vs. intrinsic rate", we calculated the space-time yield (STY). The results are shown in the Table R6. Compared to recently reported Ru-based RCF systems for generating propenyl side-chained phenols, this work exhibits competitive STY (20.3 g monomers per gram catalyst per hour) and operates at a low reaction temperature (140° C) without requiring organic solvents.

Table R6. Comparison of the space－time yield (STY) obtained in this work with recently reported Ru-based RCF systems under N_2 atmosphere.

Catalyst	Feedstock	Solvent	Temp. (° C)	Time (h)	Atmosphere	Monomers Yield (wt%)	STY (g/g/h)	Ref.
0.2Pt5Ni	birch	H_2O	140	12	N_2	38.6	20.3	This work
$Ru_{0.8}$/C	birch	methanol	235	4	N_2	35.8	22.9	1
5Ru/C	birch	110%ChCl-EG	190	2	N_2	36.2	3.8	2

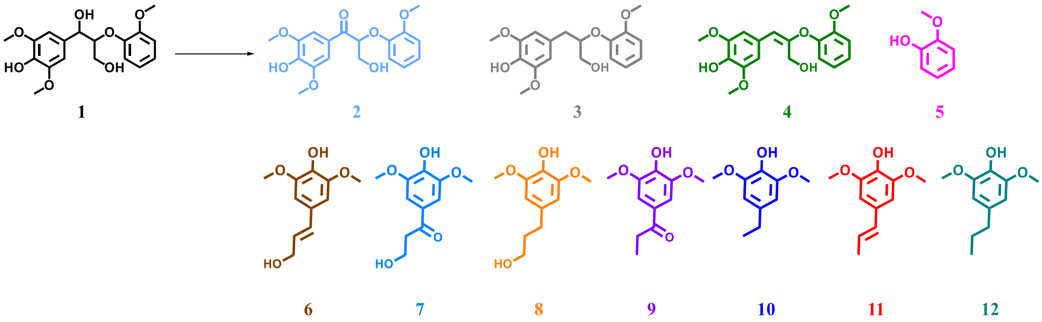
1. Huang, H., Zhang, X., Ma, L. & Liao, Y. Reductive catalytic fractionation of lignocellulose toward propyl- or propenyl-substituted monomers and mechanistic understanding. *Angew. Chem. Int. Ed.* **64**, e202502545 (2025).
2. Li, Y., *et al.* Hydrogen-Transfer Reductive Catalytic Fractionation of Lignocellulose: High Monomeric Yield with Switchable Selectivity. *Angew. Chem. Int. Ed.* **62**, e202307116 (2023).

4. Additional control experiments are required to verify the role of xylan as the hydrogen source. For example, reactions conducted under H_2 or using external hydrogen-donating solvents should be included to determine whether the formation of propenyl-substituted phenolic products is dictated primarily by the hydrogen source rather than by the intrinsic catalytic properties. Furthermore, the selectivity toward propenyl side-chained products should be quantitatively compared with literature values to convincingly demonstrate any advantage of single-atom alloy catalysts over conventional single-atom or nanoparticle catalysts for preserving unsaturated side chains.

Reply: We thank the reviewer for raising this comment. We conducted control experiments on model compound **SG** using different types of hydrogen sources, such as xylan, isopropanol, and H_2. As shown in Table R7, the results indicate that intermediate 4 remains the primary product across all hydrogen sources, suggesting a consistent reaction pathway. Notably, using xylan as the hydrogen source yields 28.5% (6+11) olefinic phenolic monomers, whereas isopropanol and H_2 produce lower yields of 23.6% (6+11) and 17.7% (6+11), respectively. This demonstrates that the formation of propenyl-substituted phenolic products is primarily determined by intrinsic catalytic properties rather than hydrogen sources. Furthermore, the mass activity (mass of propenylphenols per gram of metal) was evaluated to benchmark against state-of-the-art Ru-based catalysts. Our single-atom alloy catalyst (0.2Pt5Ni/$NiAl_2O_4$) achieved a yield of 126.3 g per gram of Pt, significantly outperforming the recently reported 5Ru/C (6.6 g/g) [2] and $Ru_{0.8}$/C (73 g/g) [1] systems. This result highlights the distinct advantage of the single-atom alloy strategy in preserving unsaturated functional group.

The relevant description has been incorporated into the revised manuscript and supporting information (Table S12).

Table R7. Reaction pathway of lignin model compound **SG** under different hydrogen sources.



Hydrogen Sources	Conv. (%)	Yield (wt%)										
		2	3	4	5	6	7	8	9	10	11	12
Xylan (0.05 g)	100	0.4	0.1	7.6	37.1	12.0	4.5	0.1	0.7	1.6	16.5	2.7
Isopropanol (0.02 g)	100	-	-	1.1	44.3	6.8	1.7	0.1	0.1	2.2	16.8	14.4
10%H_2/Ar	100	-	-	0.5	44.7	0.7	0.6	1.4	1.8	1.4	17.0	18.2

Reaction Conditions: Substrate (0.1 g), 0.2Pt5Ni catalyst (0.05 g), H_2O (10 mL), 140 °C, 2 h, 1 atm N_2 or 10%H_2/Ar.

1. Huang, H., Zhang, X., Ma, L. & Liao, Y. Reductive catalytic fractionation of lignocellulose toward propyl- or propenyl-substituted monomers and mechanistic understanding. *Angew. Chem. Int. Ed.* **64**, e202502545 (2025).
2. Li, Y., *et al.* Hydrogen-Transfer Reductive Catalytic Fractionation of Lignocellulose: High Monomeric Yield with Switchable Selectivity. *Angew. Chem. Int. Ed.* **62**, e202307116 (2023).

5. To enhance the reliability of the results, more experimental details are needed. For example, monomer yield quantitative method, GC column temperature program, representative GC chromatograms with peak assignments, GPC results for the lignin-oil.

Reply: We thank the reviewer for raising this comment.

The quantification of phenolic monomers was carried out using GC-MS (Agilent 7890A) and quantitatively analyzed by GC (Agilent 7890B) with a flame ionization detector, both quipped with HP-5 capillary columns. Tridecane was used as an internal standard for the quantification of the liquid products. Nitrogen was employed as a carrier gas, and the following operating conditions were adopted: injection temperature 250 °C, column temperature program: 50 °C (hold time 3 min), 10 °C min^{-1} to 250 °C (hold time 5 min), detection temperature of 280 °C. Response factors for the different products were determined by calibration with commercial standards or through calculations based on the effective carbon number

(ECN) theory.[3] All experiments were performed at least three times. The phenolic monomers yields were calculated based on the Klason lignin weight, calculating formula as shown below:

$$\text{Phenolic monomers (wt\%)} = \frac{\text{Mass (total monomers)}}{\text{Mass (Klason lignin)}} \times 100\%$$

$$\text{Mole yields of monomers (\%)} = \frac{\text{Mole (total monomers)}}{\text{Mole (all monomers in lignin)}} \times 100\%$$

The mass of lignin in biomass was measured by Klason method and the mole of monomers in lignin was measured by NBO method.[4]

GC-MS spectrum and GPC profiles of the lignin-derived products from SCF of birch over $0.2Pt5Ni/NiAl_2O_4$ catalyst were shown in Figure R4 and Figure R5. The weight-average (M_w) and number-average (M_n) molecular weights of lignin decreased significantly after the reaction. Specifically, M_w dropped from 4484 to 498 g/mol, while M_n declined from 2990 to 259 g/mol.

The relevant description has been incorporated into the revised manuscript and supporting information (Figure S6 and Figure S9).

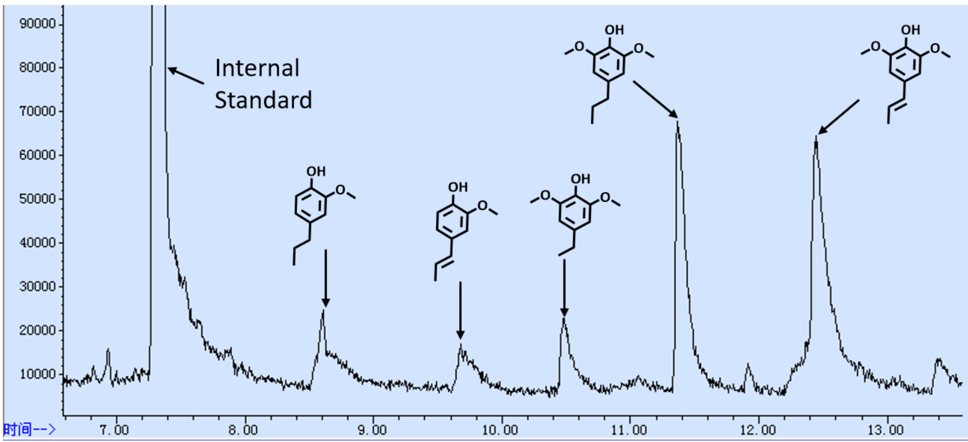


Figure R4. GC-MS spectrum of the lignin-derived monomers from birch over the $0.2Pt5Ni/NiAl_2O_4$ catalyst. Reaction conditions: birch wood (0.5 g), 0.2Pt5Ni catalyst (0.2 g), H_2O (10 mL), 140°C, N_2 1 atm, 12 h.

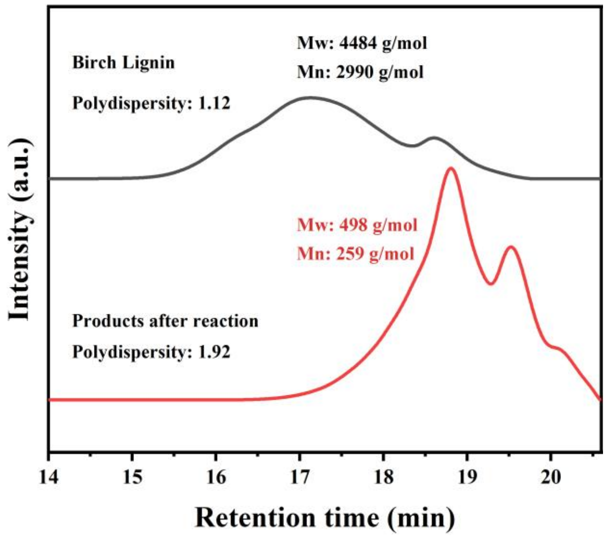


Figure R5. GPC profiles of the lignin-derived products from SCF of birch over 0.2Pt5Ni/$NiAl_2O_4$ catalyst.

3. Scanlon, J.T. & Willis, D.E. Calculation of Flame Ionization Detector Relative Response Factors Using the Effective Carbon Number Concept. *J. Chromatogr. Sci.* **23**, 333-340 (1985).
4. Shuai, L., *et al.* Formaldehyde stabilization facilitates lignin monomer production during biomass depolymerization. *Science* **354**, 329-333 (2016).

Reviewer #3 (Remarks to the Author):

Zhou et al. report the development of a single-atom alloyed Pt_1Ni catalyst for lignocellulose fractionation that enables efficient production of phenolic monomers under hydrogen-free conditions while preserving cellulose integrity. This work addresses a critical challenge in biomass valorization by achieving high monomer yields and enhanced selectivity toward unsaturated side-chain products without the need for external hydrogen or excessive metal loading, and it proposes a mechanistically well-supported catalytic strategy. Overall, the manuscript is strong and appears suitable for publication. However, several concerns should be addressed, as outlined below. Addressing these points will further strengthen the manuscript and ensure that it represents a meaningful contribution to sustainable biomass conversion and catalytic lignin valorization.

1. The authors attribute the enhanced Cα–OH activation by the oxophilicity of Ni to the higher selectivity

not clearly explained. It should be clarified.

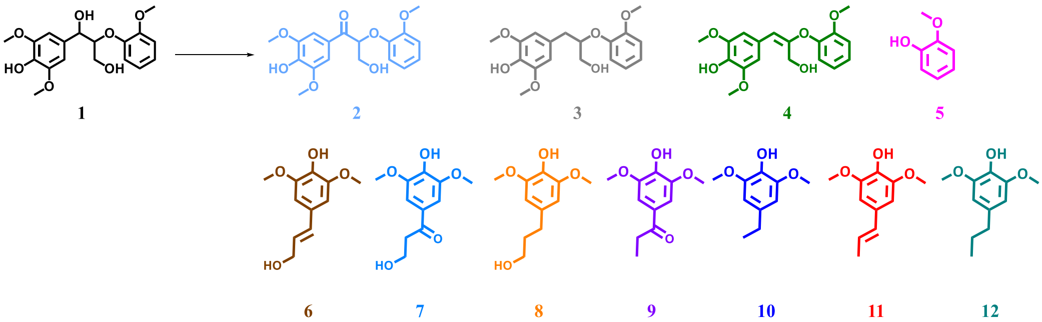
2. Self-hydrogen supplied fractionation rather than exogeneous hydrogen gas was used as the hydrogen resources in this work. Compared with direct use of hydrogen gas, whether would self-hydrogen supplied fractionation affect the reaction mechanism or the resultant selectivity?

Reply: We thank the reviewer for raising this comment. We conducted control experiments on model compound **SG** using different types of hydrogen sources, such as xylan, isopropanol, and H_2. As shown in Table R7, the results indicate that intermediate 4 remains the primary product across all hydrogen sources, suggesting a consistent reaction pathway. Notably, using xylan as the hydrogen source yields 28.5% (6+11) olefinic phenolic monomers, whereas isopropanol and H_2 produce lower yields of 23.6% (6+11) and 17.7% (6+11), respectively. This demonstrates that the formation of propenyl-substituted phenolic products is primarily determined by intrinsic catalytic properties rather than hydrogen sources.

The relevant description has been incorporated into the revised manuscript and supporting information

(Table S12).

Table R7. Reaction pathway of lignin model compound **SG** at different hydrogen sources.



Hydrogen Sources	Conv. (%)	Yield (wt%)										
		2	3	4	5	6	7	8	9	10	11	12
Xylan (0.05 g)	100	0.4	0.1	7.6	37.1	12.0	4.5	0.1	0.7	1.6	16.5	2.7
Isopropanol (0.02 g)	100	-	-	1.1	44.3	6.8	1.7	0.1	0.1	2.2	16.8	14.4
10%H_2/Ar	100	-	-	0.5	44.7	0.7	0.6	1.4	1.8	1.4	17.0	18.2

Reaction Conditions: Substrate (0.1 g), 0.2Pt5Ni catalyst (0.05 g), H_2O (10 mL), 140 °C, 2 h, 1 atm N_2 or 10%H_2/Ar.

5. The manuscript contains some typos. For example, "Tranditionally" should be corrected to "Traditionally" in page3, and "efficientcy" should be corrected to "efficiency" in page3 and page12. The authors are encouraged to carefully proofread the text and revise these issues for clarity and consistency.

Reply: We thank the reviewer for pointing out this issue. We have corrected the specified typos and conducted a careful proofread of the full manuscript and SI to improve clarity and consistency.

Response to reviewer comments

Reviewer #2 (Remarks to the Author):

The manuscript provides detailed explanations for the proposed issues and supplements relevant experiments. There are still some questions that require to provide further clarification before publication.

2) Aerobic metals Mo and W cannot form alloy structures with Pt, but still exhibit SCF activity and a certain selectivity for propenyl products compared to 0.2% Pt. Please provide a more detailed explanation to clarify this.

Reply: We thank the reviewer for this insightful comment. Although Pt does not form alloys with Mo or W, the enhanced catalytic performance can be attributed to the interfacial synergy between Pt and the oxophilic MO_x (M = Mo, W) species. Specifically, the oxophilic interface facilitates selective C–O bond cleavage, while hydrogen spillover from Pt promotes the reduction of interfacial oxygen species. Although the stronger oxophilicity of Mo and W makes the subsequent desorption of H_2O and completion of the catalytic cycle more challenging, this interfacial mechanism still enables a more efficient and selective pathway toward propenyl products compared to 0.2% Pt alone.